Journal of Substance Abuse Treatment 113 (2020) 108003

Contents lists available at ScienceDirect

Journal of Substance Abuse Treatment

journal homepage: www.elsevier.com/locate/jsat

Functioning of adults in alcohol use disorder treatment: Role of concerned others

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ARTICLE INFO

Keywords:
Alcohol use disorder
Treatment
Concerned others
Relationships
12-Step groups

ABSTRACT

Objective: This study examined patients in treatment for alcohol use disorders ("Patients") and their "concerned others" (COs—family and friends): (1) Did Patients' functioning differ according to COs' study participation? Among Patients with participating COs, (2) did Patients and COs agree on Patients' functioning, and (3) was Patients' functioning associated with COs' functioning and quality of CO-Patient relationships?
Method: Four-hundred and two Patients (mean age = 44, majority white men) and 277 COs (mean age = 52, majority white women) completed validated assessments.
Results: (1) Unexpectedly, Patients who did not identify a CO for potential study participation had more protective factors against future substance use and more readiness to participate in Alcoholics Anonymous (AA) than patients who did identify a CO. (2) Patients had higher scores than COs did when rating the Patient's protective factors, viewed the Patient-CO relationship as having more resources and fewer stressors than COs did, and reported fewer incidents of violence toward the CO than the CO did. (3) Patients had higher risk factors scores when their COs binge drank, and the Patient-CO relationship had more stressors and violence. Patients had higher protective factors scores when COs had greater readiness for Al-Anon participation, and Patients had attended more AA meetings, reported more resources in their relationship with their CO, and used more negotiation tactics when in conflict with their CO.
Conclusions: Findings suggest that interventions to improve Patients' functioning should be broadened beyond COs who are spouses or partners, decrease COs' binge drinking, facilitate 12-step participation, decrease relationship stressors and conflict, and increase relationship resources.

1. Introduction

People entering treatment for an alcohol use disorder (referred to as "Patients") often have few family members or friends in their social network to support their recovery. Years of active addiction may erode Patients' ongoing relationships with these "concerned others" (COs) (McKay, 2017). However, social connections constitute a key component of recovery (Pettersen et al., 2019).

This study examined associations between aspects of the Patient-CO relationship and Patients' functioning when Patients were in treatment. It addressed three main questions. (1) Did Patients' functioning differ in relation to COs' identification and participation in the study? (2) Among Patients with a CO who participated in the study, to what extent did Patients and COs agree on Patients' functioning? (3) Also among Patients with a participating CO, was the Patient's functioning associated with the CO's functioning and the quality of the CO-Patient relationship? Because it is important to understand characteristics of patients' social networks to promote positive treatment outcomes (Haverfield, Ilgen, Schmidt, Shelley, & Timko, 2019), the purpose of this study was to inform clinical interventions for Patients by learning more about the role of COs in Patients' functioning. Considerable evidence supports the idea that alcohol and drug use is shaped by friends

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https://doi.org/10.1016/j.jsat.2020.108003
Received 17 January 2020; Received in revised form 6 March 2020; Accepted 28 March 2020
0740-5472/ Published by Elsevier Inc.

and family members who are important socialization agents (Rulison, 2020). Thus, COs are uniquely positioned to support recovery when they are motivated to do so; COs, therefore, demand closer scrutiny (Archer, Harwood, Stevelink, Rafferty, & Greenberg, 2019).

1.1. Patients' functioning and COs' identification and participation

We compared three groups of Patients: (a) Patients who did not identify a CO (a friend or family member who was supportive of their recovery) for potential participation in the study, (b) Patients who identified a CO but the CO did not participate in the study, and (c) Patients whose identified CO participated in the study. Although study participation is not equivalent to treatment participation, COs' willingness to participate in research with their Patient may be a proxy for their willingness to cooperate with treatment (Hallgren & McCrady, 2016).

Prior research suggests variability in Patients' inclinations to elicit COs' involvement in their treatment. Regarding study participation, in research that recruited women with alcohol use disorders and their male partners, the necessity of having the male partner participate was the single largest reason given for eligible women declining to enter the study (34%) (McCrady, Epstein, Cook, Jensen, & Hildebrandt, 2009). Women who enrolled in the study were older and more likely to be married than women who did not enroll.

In a series of studies, Kidorf and colleagues examined the extent to which substance use disorder patients were willing to ask substance-free COs to participate in the treatment program to support their recovery efforts. In a study of 59 patients in opioid treatment (Kidorf et al., 2005), all patients identified a drug-free CO, and 55 (93%) brought the CO to a treatment session. In another study of 355 patients (Kidorf, Latkin, & Brooner, 2016), 98% reported having a substance-free family member (spouse/partner, sibling, adult child, parent, other relative) or friend in their social network, and 89% had at least one network member they were willing to invite to treatment. Kidorf, Brooner, Peirce, Gandotra, and Leoutsakos (2018) developed the Social Network Activation Group for patients who failed to bring a non–-substance using CO because they did not have any, or were resistant to include the ones they had in treatment. The purpose was to encourage patients to improve opportunities for securing substance-free support and resolve ambivalence about having substance-free COs included in treatment. Of 157 patients, 101 (64%) were referred to the group because they did not identify a CO; however, only 66 were retained in the group. Of the 66 patients in the intervention, 36% brought a CO to treatment, most commonly a family member.

We built upon these studies by examining whether Patients' functioning was associated with their willingness or ability to identify a supportive CO, as well as COs' willingness to participate in a research project. This analysis helps to determine whether poorer functioning by Patients is linked to their unwillingness or inability to identify a supportive CO, and to the extent that COs cooperate during Patients' treatment.

1.2. Do Patients and COs agree on Patients' functioning and their relationship?

Using data from Project MATCH (a large clinical trial of treatment matching among people with alcohol use disorders), clients' and collaterals' (individuals nominated by clients who were knowledgeable about the clients' drinking status) self-reports agreed on clients' drinking (frequency and amount) at treatment admission (Babor, Steinberg, Anton, & Del Boca, 2000). Among men in treatment for alcohol use disorders, self-reported frequency of illicit drug use was highly correlated with their female partners' reports of the client's drug use for the most commonly used drugs of cannabis, cocaine, and heroin (O'Farrell, Fals-Stewart, & Murphy, 2003). These results suggest that Patients and COs may generally agree on amounts of Patients' substance use at the time of treatment admission. However, studies have not addressed the extent to which Patients and COs agree on the Patient's risk and protective factors with regard to continued recovery. Discrepant views of the Patient's recovery may detract from the dyad's connection and mutual encouragement, limiting opportunities for the Patient to receive potentially helpful social support.

1.3. Associations between Patients' and COs' functioning

To examine associations between Patients' and COs' functioning, this study drew from a conceptual model integrating a stress and coping perspective on recovery from alcohol use disorders (Finney & Moos, 1995; Moos, Brennan, Schutte, & Moos, 2010; Moos, Finney, & Cronkite, 1990). The model identifies sets of variables that determine Patients' outcomes, including their demographic characteristics; substance use severity; COs' functioning; and stressors, resources, and conflict in Patient-CO relationships. The model incorporates the mutual influence of Patients' functioning and its determinants over time.

Social factors have a robust impact on drinking outcomes. For example, more family cohesion was associated with a lower prevalence of alcohol use disorder among a random sample of adults in San Juan, Puerto Rico (Caetano, Vaeth, & Canino, 2017), and having greater social support was linked to recovery from harmful or hazardous drinking among a nationally representative sample of veterans (Fuehrlein et al., 2018). A systematic review of factors associated with alcohol use disorder relapse (Sliedrecht, de Waart, Witkiewitz, & Roozen, 2019) concluded that a positive social context and nondrinking social support are protective, whereas heavy drinking among social network members is associated with greater relapse risk.

Studies that examined more specific aspects of Patients' social context agree that when COs are better functioning and have a higher quality relationship (e.g., fewer stressors) with Patients, Patients have better drinking outcomes. For example, when COs consumed less alcohol and experienced fewer physical symptoms and less depression, Patients had better drinking-related outcomes over a 10-year period (Moos et al., 1990). A survey of Alcoholics Anonymous (AA) members found significant positive associations of duration of Al-Anon attendance by a CO with the Patient's duration of AA attendance, and with lower stress reported for Patients and COs (McBride, 1991). COs who received psychotherapy focused on Al-Anon facilitation (Nowinski, 1999) had decreased depression, and the patient had reduced alcohol consumption at one-year follow-up (Rychtarik & McGillicuddy, 2005). Among individuals with alcohol use disorders followed for one year, abstinence was associated with better family relationships (more cohesion and expression and less conflict) (Dale et al., 2017), and among those followed for five years, abstinence was associated with high levels of friend support (McCutcheon, Lessov-Schlaggar, Steinley, & Bucholz, 2014).

1.4. Current study

This study collected data from 402 Patients and 277 COs of these Patients when the Patients were entering alcohol use disorder treatment. Hypotheses were:

(1) Patients whose CO participated in the study would have better functioning than Patients without a participating CO.
(2) Patients and COs would not differ on an assessment of Patients' substance use. We did not form hypotheses on their views of Patients' risk and protective factors for future substance use due to lack of previous studies on this topic.
(3) Generally, significant positive associations would be found between COs' and Patients' better functioning. Again, we did not form more specific hypotheses due to lack of guidance from previous studies.

Findings will be useful to identify factors associated with Patients'

functioning, such as aspects of the Patient-CO relationship, that treatment programs' interventions could modify to help improve Patients' outcomes.

2. Material and methods

2.1. Sample and procedure

2.1.1. Patients

The sample comprised 402 patients entering residential treatment for alcohol use disorder. To accrue the sample, first, patients were asked for permission by treatment staff to be contacted by the study team. Patients who agreed were contacted in person at the treatment program by study staff, and were screened (if they agreed) for study eligibility (i.e., at least 19 years old, could communicate in English, did not have a conservator). Of 453 patients approached about the study, 94% (n = 426) agreed to screening for eligibility, and of these, 96% (n = 409) were eligible for the study. Of eligible patients, 402 (98%) were enrolled and completed the study's assessment, for which they were given $25. The only reason for ineligibility was having a conservator. A conservator is appointed by a court to manage an individual's financial affairs due to physical or mental limitations or old age; the conservator is required to have accountings regularly approved by the court, and may be removed if no longer needed upon petition or failure to perform duties. Patients who were screened and eligible were asked to provide informed consent.

The Patient sample comprised mainly men (75.3%); most were not married (82.5%) and were unemployed (83.5%). The majority were white (67.6%); 23.7% were black, and 12.8% were Hispanic or Latino. The mean age was 44.0 years (SD = 12.6; range = 19–75), the mean years of education was 13.0 (SD = 2.0), and the mean annual income was $24,710 (SD = $26,419).

2.1.2. Concerned others

The sample also included 277 COs. Participants in the Patient sample were asked to provide a signed release naming potential COs (i.e., someone at least 19 years old who is important to and supportive of the patient and the current treatment episode) and COs' contact information. The study team attempted to contact a CO for each patient when a CO was identified to obtain informed consent and to ask them to complete the study assessment.

Procedures to recruit COs included: (a) mail the CO a letter that introduced the study and gave the CO instruction on how to contact study staff (to decline further contact or obtain more information); and (b) about a week later, call the CO to discuss study participation and arrange an appointment for enrollment. Prospective COs were called twice a week for up to five weeks until they responded (or did not respond). Calls included all of an individual's phone numbers (landlines and cell phones) and leaving a message on every third call for a total of up to three messages. COs who were not reached after three weeks of calls were sent a second letter. Of 413 COs approached about the study, 277 (67%) were enrolled and completed the study assessment, for which they were given $25. Reasons for nonenrollment (n = 136) included: the CO did not respond to contact attempts (59%), the CO declined participation when contacted (27%), and the CO did not attend scheduled enrollment sessions (14%).

The CO sample comprised mainly women (77.3%); most were not married (58.5%) and most were employed (57.7%). The majority were white (68.3%); 21.2% were black, and 9.9% were Hispanic or Latino. The mean age was 51.9 years old (SD = 14.8; range = 19–89), the mean years of education was 13.8 (SD = 2.0), and the mean annual income was $42,300 (SD = $41,100). The CO's relationship to the patient was: current spouse or partner (30.1%), former spouse or partner (2.6%), parent (25.0%), sibling (13.2%), offspring (11.4%), other relative (2.6%), or friend (15.1%).

All study procedures were Institutional Review Board compliant.

2.2. Measures: patients

2.2.1. Alcohol use disorder

Patients rated their alcohol use disorder on the Brief Addiction Monitor (BAM; Cacciola et al., 2013). The BAM yields three subscales referring to the 30 days before treatment admission (see Table 1b for sample means and standard deviations). (1) Substance Use is the sum of three items, e.g., number of days drank alcohol, number of days used any illegal/street drugs or took prescription medication not as prescribed or without prescription (for each, 0 = 0, to 4 = 16–30 days); scores range from 0 to 12. (2) Risk Factors for substance use is the sum of six items, e.g., physical health (0 = excellent, 4 = poor); in any situations that or with any people who might increase risk for using alcohol or drugs (0 = 0, 4 = 16–30 days); scores range from 0 to 24. (3) Protective Factors is the sum of six items, e.g., confidence in ability to be completely abstinent from alcohol and drugs in the next 30 days, religion or spirituality helps support recovery (0 = not at all, 4 = extremely); scores range from 0 to 24.

The BAM has been used in prior studies to measure outcomes among clients receiving substance use disorder treatment. Research using the BAM has shown, for example, that severity of substance use was reduced among clients in a joint substance use–pain treatment program (Stockin, Wandner, Kurihara, Spevak, & Griffith, 2019) or for those receiving intervention for co-occurring substance use-PTSD (Najavits, Lande, Gragnani, Isenstein, & Schmitz, 2016). It also showed a consistent trend of decreased use and risk factors and increased protective factors among substance use disorder clients treated with contingency management (Ruan, Bullock, & Reger, 2017). Further, a randomized controlled trial found that detoxification clients assigned to telephone monitoring improved more on BAM subscales than those assigned to usual care (Timko et al., 2019). Another study found no differences on the BAM between residential substance use clients randomly assigned to a feedback or control condition, although the BAM was reliable (Blonigen, Timko, Jacob, & Moos, 2015).

2.2.2. Treatment and AA

Patients reported the number of times (episodes, not number of days or sessions) they had been treated for alcohol use; this included inpatient, residential, and outpatient treatment. They also reported the number of AA meetings they had attended during the past six months. Further, they completed a 5-item scale measuring AA readiness. Items (e.g., work the 12 steps) were rated on a 4-point scale (1 = not ready to do; 4 = already doing) and averaged (Cronbach's alpha = 0.89). See Table 1b for sample means and standard deviations.

2.2.3. Patient-CO relationship

The Patient-CO relationship was assessed using the Health and Daily Living Form (HDL; Moos, Cronkite, & Finney, 1992). The HDL is psychometrically sound and has been used extensively (Bi, Moos, Timko, & Cronkite, 2015; Holahan et al., 2018). Patients rated their relationship with their CO on the HDL on 10 items with regard to how often events occurred (0 = never, 4 = very often; see Table 2 for sample means and standard deviations). *Relationship stressors* was the sum of five items, e.g., disagreed about important things; alpha = 0.85. *Relationship resources* was the sum of five items, e.g., calmly discussed something together; alpha = 0.83.

2.2.4. Patient-CO conflict

Patient-CO conflict was assessed using an adaptation of the Conflict Tactics Scale Short Form (CTS2; Straus & Douglas, 2004), which measures three tactics used in conflicts between dyads. Tactics were assessed for the past six months; see Table 2 for means and standard deviations. Use of *negotiation* was the sum of two items (e.g., Explained my side or suggested a compromise for a disagreement with my CO), and asked for the number of times the Patient used this tactic. Use of *violence* was the sum of five items (e.g., I pushed, shoved, or slapped my

CO), and asked for the number of times the Patient used this tactic. *Injury* was the sum of two items (e.g., My CO had a sprain, bruise, or small cut or felt pain the next day because of a fight with me), and asked for the number of times the Patient injured the CO during a conflict.

2.3. Measures: COs

2.3.1. Substance use

COs answered the number of times they binge drank in the past 30 days (specifically, "How many times did you drink 5 or more [if you are a man] or 4 or more [if you are a woman] drinks on a single occasion"), and how many days in the past 30 days they had used any nonprescribed drugs, such as, drugs prescribed for someone else, marijuana or cannabis, or cocaine (not including over-the-counter drugs like aspirin or antacids). See Table 3 for sample means and standard deviations.

2.3.2. Patients' alcohol use disorder

COs rated their Patient's alcohol use disorder on the BAM. This yielded the same three subscale scores as for Patients referring to the 30 days before treatment admission: substance use, risk factors, and protective factors (see Table 2).

2.3.3. Patient-CO relationship

COs rated their relationship with their Patient to yield the same two subscale scores on the HDL of stressors (alpha = 0.89) and resources (alpha = 0.89); see Table 2.

2.3.4. Patient-CO conflict

Patient-CO conflict was assessed by COs using the adapted CTS2 to yield three scores: *negotiation* (e.g., my Patient explained his or her side or suggested a compromise for a disagreement with me); *violence* (e.g., My patient pushed, shoved, or slapped me), and *injury* (e.g., I had a sprain, bruise, or small cut or felt pain the next day because of a fight with my Patient); see Table 3.

2.3.5. CO functioning

CO functioning was assessed in three domains using the HDL (see Table 3). Items for *quality of life* and for *mental health* were rated on a 5-point scale (0 = never, 4 = very often). Quality of life was the mean of 10 items (e.g., On the whole, how often have you been satisfied with your overall quality of life and well-being); alpha = 0.76. Mental health was the mean of 10 items (e.g., How often have you experienced feeling hopeful); negative symptoms (e.g., anxious) were reverse scored (alpha = 0.82). General health was one item (In general, how would you describe your health) rated on a 4-point scale (1 = excellent, 2 = good, 3 = fair, 4 = poor).

2.3.6. CO coping

COs' coping was assessed in the HDL (see Table 3). The CO rated items with respect to their Patient's drinking on a scale from 0 (definitely no) to 3 (definitely yes). Approach coping was the mean of 24 items (alpha = 0.87), e.g., Did you think of different ways to deal with the problem; Did you make a plan of action and follow it? Avoidance coping was the mean of 24 items (alpha = 0.83), e.g., Did you try not to think about the problem? Did you think that the outcome will be decided by fate?

2.3.7. Al-Anon

COs reported on the number of Al-Anon meetings they had attended in the past six months. They also completed a 5-item scale measuring readiness to participate in Al-Anon. Items (e.g., work the 12 steps) were rated on a 4-point scale (1 = not ready to do; 4 = already doing) and then averaged (alpha = 0.90). See Table 3 for sample means and standard deviations.

2.4. Analysis plan

2.4.1. Question 1

For Question 1 (is the Patient's functioning associated with a CO's participation in the study), Patients' functioning was operationalized as Patients' scores on the three BAM subscales of substance use, risk factors, and protective factors. We first conducted analyses of variance to identify any significant differences in demographic characteristics among three groups of Patients: did not identify a CO, identified a CO but the CO did not participate in the study, and identified a CO who participated. Then, we conducted analyses of covariance (ANCOVAs) that controlled for demographic differences to compare the three groups on Patients' functioning and treatment and AA participation.

2.4.2. Question 2

For Question 2 (do Patients and COs agree on Patients' functioning and the Patient-CO relationship), we conducted ANCOVAs that controlled for demographic differences to compare Patients and COs. Patients and COs were compared on Patients' functioning (i.e., the three BAM subscales) and the Patient-CO relationship (i.e., resources, stressors) and conflict (i.e., negotiation, violence, injury).

2.4.3. Question 3

For Question 3 (is Patients' functioning associated with COs' functioning and the Patient-CO relationship), we conducted multiple regressions for each indicator of Patient functioning; all regressions controlled for Patients' demographics. First, we examined associations of COs' reports (i.e., CO substance use, functioning, coping, relationship and conflict with Patient, and Al-Anon participation) or Patients' reports (treatment and AA participation, relationship and conflict with CO) with each indicator of Patient functioning. Next, for each functioning indicator, a final regression entered each of the significant predictors identified across the CO and Patient regression analyses to determine the final, most parsimonious model predicting that aspect of Patient functioning. Prior to conducting the final regressions, we examined the intercorrelations of all predictors to avoid the problem of multicollinearity. Patients' relationship stressors and relationship resources had a correlation of −0.50. Otherwise, intercorrelations ranged from −0.17 to 0.35. In addition, each variance inflation factor was < 2 (Chatterjee & Simonoff, 2013). Thus, problematic multicollinearity was not present. Overall, this analytic method for question 3 was used to enable parsimonious explanatory models of Patients' functioning (Bursac, Gauss, Williams, & Hosmer, 2008; Hastie, Tibshirani, & Friedman, 2009).

3. Results

3.1. Is Patients' functioning associated with COs' participation?

Of the participating Patients (n = 402), 277 (69%) identified a CO who participated in the study, 74 (18%) identified a CO but the CO did not participate, and 51 (13%) were unable or unwilling to name a CO. Comparisons of the three groups of Patients on baseline demographic characteristics showed that Patients who did not identify a CO were more likely to be women (Table 1a). And, Patients who identified a CO who did not participate were less likely to be employed and were older.

ANCOVAs compared the three groups on their functioning and treatment and AA participation, controlling for Patients' gender, age, and employment status (Table 1b). Patients who did not identify a CO scored higher on protective factors and reported more readiness to participate in AA.

3.2. Do Patients and COs agree on Patients' functioning?

Because correlations of Patients' demographic factors (listed in Table 1a) with Patients' functioning found older age to be associated

Table 1a
Demographic characteristics of three groups of Patients: CO was identified and participated (n = 277), CO was identified and did not participate (n = 74), and CO was not identified (n = 51).

Patients' demographics	Identified and participated M (SD) or % (n)	Identified and did not participate M (SD) or % (n)	Not identified M (SD), % (n)	F/X^2 (p)
Male	77.7 (215)	78.3 (58)	54.5 (28)	10.17 (0.006)
White	67.7 (187)	67.4 (50)	67.4 (34)	0.00 (0.998)
Married	18.9 (52)	12.0 (9)	20.5 (10)	2.78 (0.249)
Employed	19.0 (53)	7.6 (6)	20.5 (10)	7.99 (0.018)
Age	43.4[a] (12.5)	47.1[a,b] (13.4)	40.5[b] (10.4)	4.86 (0.008)
Income ($)	25,556 (26,084)	22,477 (28,647)	24,072 (23,960)	0.46 (0.634)

Note: Means that share a superscript are significantly different.

with lower risk factors scores (r = −0.16, p < .01), Patient age was a covariate in comparisons of Patients and COs on Patients' functioning and the Patient-CO relationship and conflict. As shown on Table 2, Patients and COs did not differ on their assessments of the Patient's substance use or risk factors. Patients had higher scores than COs did on the Patient's protective factors. Patients also viewed the Patient-CO relationship as having more resources and fewer stressors than COs did. Further, Patients reported fewer incidents of violence toward the CO than the CO reported.

3.3. Is Patients' functioning related to COs' functioning and the relationship?

3.3.1. CO-reported functioning and relationship with Patient

In addition to Patient age, in regression analyses we entered each set of CO measures (substance use; functioning; coping; relationship with Patient; conflict with Patient; and Al-Anon participation) to predict Patients' BAM scores on the three indicators of Patients' functioning (substance use, risk factors, and protective factors). COs' binge drinking on more days was associated with Patients' reports of more Risk Factors for continued substance use (Table 3). In addition, COs' greater readiness to participate in Al-Anon was associated with Patients' reports of having more protective factors. Otherwise, COs' functioning, coping, relationship and conflict with the Patient, and Al-Anon attendance were not related to Patients' functioning.

3.3.2. Patient-reported functioning and relationship with CO

In regressions for which we entered each set of Patient measures (treatment and AA, Patient-CO relationship, and Patient-CO conflict), we controlled for Patient age. Patients with a history of more alcohol treatments scored higher on the BAM's substance use and risk factors subscales, and lower on the protective factors subscale (Table 4). Patients with a history of attending more AA meetings scored lower on the substance use subscale and higher on the protective factors subscale.

Patients reporting more resources in their relationship with their CO also reported having more protective factors. In contrast, Patients reporting more relationship stressors reported more substance use and more risk factors. Similarly, Patients reporting use of more negotiation when in conflict with their COs reported more protective factors.

Table 2
ANCOVAs comparing Patients and COs (n = 277 pairs) on Patients' functioning and the Patient-CO relationship and conflict (controlling for Patient age).

	Patient M (SD)	Concerned other M (SD)	F (p)
Patient functioning			
Substance use	6.7 (4.1)	6.4 (3.7)	1.10 (0.295)
Risk factors	15.6 (5.6)	16.1 (5.8)	0.26 (0.607)
Protective factors	13.2 (4.1)	10.0 (4.2)	69.03 (< 0.001)
Patient-CO relationship			
Resources	17.4 (4.9)	13.1 (5.7)	70.34 (< 0.001)
Stressors	7.9 (5.3)	11.2 (5.9)	32.80 (< 0.001)
Patient-CO conflict			
Negotiation	42.1 (65.7)	45.2 (117.1)	0.116 (0.734)
Violence	12.6 (29.4)	31.9 (141.9)	4.72 (0.030)
Injury	0.2 (1.4)	0.5 (5.1)	0.86 (0.354)

Patients reporting use of more violence had more substance use and risk factors.

3.3.3. COs' and Patients' predictors of Patients' functioning

We conducted regressions to predict each aspect of Patients' functioning on the BAM subscales from the significant predictors yielded from the previous regressions (shown in Tables 3 and 4). The results of the final regressions are presented in Table 5.

Patients' substance use scores were predicted by Patients having a history of more alcohol treatments, having attended fewer AA meetings, and having more stressors in their relationship with their CO. The model explained 9% of the variance (adjusted R^2 = 8%, F [4272] = 6.39, p ≤ .001) in this outcome.

Patients' risk factors were predicted by Patients having COs who binge drank more often, and by more stressors and violence in the Patient-CO relationship. The model explained 22% of the variance (adjusted R^2 = 18%, F[4272] = 6.78, p < .001) in this outcome.

Patients' protective factors were predicted by Patients' COs having more readiness to participate in Al-Anon, as well as by Patients' own attendance of more AA meetings. In addition, higher protective factors

Table 1b
Analyses of covariance comparing three groups of Patients (controlling for Patients' gender, age, and employment status) on CO study participation.

	Identified and participated M (SD)	Identified and did not participate M (SD)	Not identified M (SD)	F (p)	Overall sample M (SD)
Patients' functioning					
Substance use	6.8 (4.1)	6.8 (4.2)	7.8 (3.3)	1.18 (0.309)	6.9 (4.0)
Risk factors	15.5 (5.6)	16.7 (5.6)	16.2 (4.9)	1.71 (0.182)	15.9 (5.5)
Protective factors	13.2 (4.1)	11.9 (3.9)[a]	14.2 (3.9)[a]	4.76 (0.009)	13.0 (4.1)
Patients' treatment and AA					
No. alcohol treatments	6.0 (9.1)	5.6 (9.5)	4.9 (5.6)	0.31 (0.730)	5.8 (8.9)
No. AA meetings attended	25.8 (33.9)	22.1 (31.7)	23.8 (28.3)	0.42 (0.658)	24.7 (32.8)
AA readiness	3.0[a] (0.8)	3.1 (0.7)	3.4[a] (0.5)	5.18 (0.006)	3.1 (0.8)

Note: Means that share a superscript are significantly different.

Table 3
Regressions predicting Patients' functioning from COs' reports of COs' functioning and the CO-Patient relationship (Patient age is controlled).

CO functioning	Patients' substance use b (p)	Patients' risk factors b (p)	Patients' protective factors b (p)	M (SD)
Binge drank	0.155 (0.056)	0.160 (0.046)	−0.080 (0.327)	1.1 (3.9)
Used non-prescribed drugs	−0.005 (0.955)	0.013 (0.873)	0.082 (0.320)	0.8 (4.2)
Quality of life	−0.017 (0.812)	−0.021 (0.773)	−0.062 (0.389)	2.9 (0.6)
Mental health	0.055 (0.447)	0.004 (0.952)	0.007 (0.921)	2.3 (0.7)
Health	−0.044 (0.512)	−0.065 (0.326)	0.097 (0.145)	2.1 (0.7)
Approach coping	0.000 (0.997)	−0.069 (0.344)	0.026 (0.724)	1.8 (0.6)
Avoidance coping	0.022 (0.764)	−0.032 (0.662)	−0.008 (0.910)	1.2 (0.5)
Relationship resources	−0.006 (0.929)	−0.102 (0.141)	0.116 (0.100)	13.1 (5.7)
Relationship stressors	0.090 (0.212)	0.005 (0.942)	0.020 (0.784)	11.2 (5.9)
Patient's negotiation	0.092 (0.159)	0.040 (0.533)	−0.018 (0.785)	45.7 (116.6)
Patient's violence	0.039 (0.557)	0.095 (0.142)	−0.004 (0.952)	31.9 (141.3)
Patient's injury	0.088 (0.164)	0.070 (0.262)	−0.082 (0.201)	0.5 (5.0)
No. Al-Anon meetings	−0.002 (0.971)	0.013 (0.837)	0.077 (0.210)	1.8 (7.6)
Al-Anon readiness	−0.094 (0.131)	−0.055 (0.368)	0.198 (0.001)	3.1 (0.8)

Table 4
Regressions predicting Patients' functioning from Patients' reports of Patients' treatment and 12-step participation and the Patient-CO relationship (Patient age is controlled).

	Patients' substance use b (p)	Patients' risk factors b (p)	Patients' protective factors b (p)
No. alcohol treatments	0.141 (0.006)	0.140 (0.006)	−0.101 (0.046)
No. AA meetings	−0.165 (0.001)	−0.076 (0.132)	0.233 (0.000)
AA readiness	0.012 (0.814)	−0.044 (0.376)	−0.035 (0.480)
Relationship resources	0.094 (0.132)	−0.020 (0.740)	0.235 (0.000)
Relationship stressors	0.262 (0.000)	0.345 (0.000)	−0.038 (0.546)
Patient's negotiation	0.001 (0.982)	0.051 (0.370)	0.150 (0.010)
Patient's violence	0.129 (0.032)	0.212 (0.000)	−0.031 (0.601)
Patient's injury	0.030 (0.604)	0.046 (0.403)	−0.068 (0.229)

scores were predicted by Patients reporting more resources in their relationship with their CO, and using more negotiation tactics when in conflict with their CO. The model explained 13% of the variance (adjusted R^2 = 11%, F[5271] = 7.64, p ≤ .001) in this outcome.

4. Discussion

This study of 402 Patients in treatment for their alcohol use disorder, and a subset of 277 Patients and their COs, addressed three main questions. It found that Patients who did not identify a CO for the study did not (contrary to our hypothesis) have poorer functioning and may not have been lacking in social support. In addition, participating COs had more negative views of the Patient and the Patient-CO relationship than Patients did. Further, COs' own drinking and willingness to participate in the 12-step Al-Anon program, and Patients' views of their relationships with their COs, were related to Patients' functioning. Below, we expand on these findings and their implications for interventions delivered during treatment to Patients and COs to improve outcomes of both members of the dyad.

4.1. Patients' functioning and COs' participation in research

This study found that the majority of Patients identified a CO, and that the majority of COs approached about the study participated in it. We hypothesized that not identifying a CO would be associated with Patients having more severe disorders. We did so because more severe and chronic substance use histories are often associated with estrangement from important relationships (Pettersen et al., 2019). However, this study found that Patients who did not identify a CO for potential study participation had more protective factors against future substance use, and reported more readiness to participate in AA. Thus not identifying a CO did not appear to indicate an absence of social support. Rather, Patients without a CO were likely experiencing or anticipating the receipt of support through AA participation. Unfortunately, we did not document specific reasons that Patients did not identify a CO. Their reasons, such as not having any CO, or having a CO but not wanting to burden them in any way (Kidorf et al., 2018), may be related to Patients' experience of support and to treatment outcomes. Subsequent research should examine how Patients' reasons are related to support and outcomes.

4.2. Patients' and COs' agreement on Patients' functioning

As previous studies have found (Babor et al., 2000; O'Farrell et al., 2003) and as we hypothesized, Patients and COs agreed on their assessments of Patients' substance use. They also agreed on assessments of the Patients' risk factors for continued substance use. However, COs saw Patients as being less protected against continued substance use, and viewed the Patient-CO relationship as having fewer resources and more stressors, and being more violent. The COs' negative views on Patients'

Table 5
Regressions predicting Patients' functioning from COs' and Patients' reports (Patient age is controlled).

	Patients' substance use b (p)	Patients' risk factors b (p)	Patients' protective factors b (p)
CO binge drank	–	0.173 (0.032)	–
CO Al-Anon readiness	–	–	0.124 (0.028)
Patient no. alcohol treatments	0.136 (0.014)	0.142 (0.085)	−0.078 (0.160)
Patient no. AA meetings	−0.145 (0.008)	–	0.155 (0.006)
Patient-reported relationship resources	–	–	0.185 (0.001)
Patient-reported relationship stressors	0.169 (0.004)	0.301 (0.001)	–
Patient-reported patient's negotiation	–	–	0.111 (0.039)
Patient-reported patient's violence	0.080 (0.170)	0.173 (0.032)	–

protective factors may be related to Patients already having a mean of 5.8 (SD = 8.9) previous treatment episodes. This is similar to findings that U.S. adults, who had resolved an alcohol or drug problem, had undertaken a mean of 5.3 serious attempts (SD = 13.4) (Kelly, Greene, Bergman, White, & Hoeppner, 2019). Patients' views of their own self-efficacy represented by the protective factors subscale are important because self-efficacy is related to better long-term treatment outcomes (Müller, Znoj, & Moggi, 2019).

In contrast to COs, Patients may be exposed to a foundation for feeling protected, self-efficacious, and hopeful when they enter treatment (Owens, Kirouac, Hagler, Rowell, & Williams, 2018). For example, Patients observe and interact with treatment staff and peers who are further along in their recovery despite numerous previous treatment episodes, and treatment staff use therapeutic techniques that reinforce the patient's decision to enter treatment. Patients may also experience more hope because entering treatment inspires a feeling of taking back responsibility for their lives (Vatne & Nåden, 2018), whereas COs may see entering treatment as another brief respite from substance use before another relapse. It could be helpful for both Patients and COs if treatment programs were to systematically provide COs with reasons that treatment initiation, even if repeated, may protect the Patient from future harm.

4.3. Determinants of Patients' functioning

We found limited evidence that COs' well-being and views of their relationship with the Patient were related to Patients' functioning. COs' reports of their own greater engagement in binge drinking predicted Patients reporting more risk factors for continued substance use. This finding is consistent with those of previous studies in which drinking among social network members was associated with poorer drinking outcomes (Sliedrecht et al., 2019). In addition, we found that COs' reports of being ready to participate in Al-Anon were associated with Patients having more protective factors. A CO's openness to working on recovery from the hardships of being involved with a Patient may encourage Patients' optimism about the chances of their own recovery. Results suggest that interventions with COs to reduce their binge drinking and facilitate their participation in Al-Anon (Nowinski, 1999) may reduce Patients' risk factors and increase their protective factors; this possibility needs to be tested in future longitudinal studies of CO-Patient dyads.

When examining Patients' functioning, we also found that when Patients reported more stressors in their relationship with their CO, they had more severe substance use and more risk factors, but when Patients reported more relationship resources, they reported more protective factors. As measured here, relationship stressors consisted of, for example, Patients disagreeing with their CO about important things, being critical or disapproving, getting angry or losing their temper, and expecting too much of their CO. Resources consisted of the Patient and CO having calm discussions, stimulating exchanges of ideas, and good times together; showing consideration and interest; and cooperating with things asked of them. Our results are consistent with those of Cranford, Tennen, and Zucker (2015), who found that marital stressors were associated with more frequent intoxication among men and women with alcohol use disorders. We also found that the use of more negotiation when in conflict with their CO and more participation in AA were associated with Patients' reports of protection against future substance use. Thus, to improve Patients' functioning, treatments might include modules on how to reduce specific relationship stressors (e.g., criticism, disapproval, and underappreciation) and violence, while increasing relationship resources (e.g., how to be considerate and how to listen attentively), conflict negotiation, and AA participation.

To improve Patients' functioning, treatment could draw from existing evidence-based approaches that target the Patient-CO relationship. Specifically, to improve drinking among patients in treatment, alcohol behavioral couples therapy (ABCT) focuses partly on enhancing the Patient-CO relationship by increasing relationship resources and reducing stressors (McCrady et al., 2019). Efforts to improve the CO-Patient relationship may require educating the Patient about the CO's negative views of the CO-Patient relationship, and the COs' lack of optimism about the Patient's self-efficacy to abstain from substance use despite treatment. So far, ABCT has been evaluated mainly among heterosexual spouses and romantic partners. Therefore, other types of Patient-CO dyads, including pairs of friends, offspring (e.g., Patient and son), and parents (e.g., Patient and mother), which made up the majority of the relationships in this study, should be studied longitudinally. In this regard, COs who were parents had higher rates of Patients entering substance use treatment than Patients with other CO relationships, possibly because parents were less worried about retaliation through violence (Archer et al., 2019).

In addition to changing Patients' behaviors that constitute relationship resources, stressors, and conflict tactics, Patients' treatment plans could focus on breaking or decoupling the link between relationship stressors and conflicts and Patients' poor functioning (Hsueh, McCormick, Merrillees, Chou, & Cummings, 2018). More generally, a recent review of family communication about adult family members' mental health problems (Gammage & Nolte, 2020) suggested that when COs (spouses, parents, siblings, children, or other relatives) learn how to stay connected with the person with the mental health problem rather than focusing on their disorder (i.e., maintain the personal and not just the caregiving relationship), they are better able to make sense of distressing behaviors, retain a sense of control, assimilate new knowledge about the disorder, and stay positive about the relationship. The review recommended clinical approaches that build COs' confidence and disorder literacy, as well as CO-Patient communication. In addition, adaptations of interpersonal psychotherapy that focus on relational aspects when treating substance use disorders (Brache, 2012) or adaptations of the interpersonal effectiveness skills-building component within dialectical behavior therapy (Dimeff & Linehan, 2008; May, Richardi, & Barth, 2016) may also be helpful for Patients attempting to achieve less chaotic and more stable relationships with their COs.

Regarding AA participation, for most individuals seeking help for alcohol problems, increasing AA attendance leads to short- and long-term decreases in alcohol consumption that cannot be attributed to self-selection (Humphreys, Blodgett, & Wagner, 2014). Studies of methods to facilitate AA participation have found interventions to be effective at increasing AA attendance and involvement, and to be associated with better drinking and other outcomes (Kaskutas, Subbaraman, Witbrodt, & Zemore, 2009; Timko, Debenedetti, & Billow, 2006; Vederhus, Timko, Kristensen, Hjemdahl, & Clausen, 2014). Although we did not include participation in non-12-step support groups (e.g., SMART Recovery) in this study due to low utilization (only two Patients reported using these groups), they are also helpful in resolving alcohol use disorders (Zemore, Lui, Mericle, Hemberg, & Kaskutas, 2018). Participation in AA or other support groups may help Patients to learn more about the strategies for and benefits of reducing stressors and violence, and increasing resources and use of conflict negotiation in their relationships with COs. Learning may occur through sponsorship or simple observations of peers with longer durations of recovery.

4.4. Limitations

Associations between Patients' functioning and its predictors were cross-sectional, such that causal conclusions should not be drawn. Future longitudinal studies of Patients' and COs' functioning need to examine the extent to which their baseline substance use, health, and other functioning severity explains subsequent outcomes. Second, sampling methods limit the generalizability of results. Our sample included Patients who had entered treatment and their COs; therefore, results may not be replicated among Patients who do not choose to obtain treatment or are unable to access it. Third, because we examined three broad questions, we conducted multiple analyses, increasing the

chances of spurious results (Type 1 error). In addition, in the regression analyses to predict Patients' functioning, relatively small amounts of variance were explained in the dependent variables. However, significant results obtained were conceptually sound and clinically meaningful (Moos et al., 2010).

5. Conclusion

This study examined the role of COs in the functioning of women and men in treatment for alcohol use disorders. It concluded that patients who did not identify a CO for the study were not functioning more poorly than those who did, and may even have been functioning better in terms of factors related to self-efficacy. Possibly, our finding of COs' negative views of the Patient relative to Patients' views of themselves helps to explain this result; that is, Patients who did not identify a CO may be at least temporarily freed from taking on the additional task of repairing relationships with COs while focusing on other aspects of treatment progress. This study also identified potential intervention targets for COs of people in treatment for alcohol use disorders (reduce COs' binge drinking and facilitate Al-Anon participation) to improve Patients' outcomes, and for Patients themselves (learn how to reduce close relationship stressors and handle them better when they occur). Research needs to longitudinally examine these targets among heterogeneous samples of COs that include romantic partners, adult offspring, other relatives, friends, and AA peers.

CRediT authorship contribution statement

Christine Timko: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing - original draft. **Kathleen M. Grant:** Conceptualization, Investigation, Project administration, Resources, Supervision, Writing - review & editing. **Rakshitha Mohankumar:** Formal analysis, Writing - review & editing. **Michael A. Cucciare:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Writing - review & editing.

Acknowledgments

This research was supported by NIH/NIAAA (R01 AA024136 01A1 to Drs. Timko and Cucciare) and the Department of Veterans Affairs (VA), Health Services Research and Development (HSR&D) Service (RCS 00-001 to Dr. Timko). The views expressed are the authors'. No conflicts of interest are reported by any of the authors listed on this manuscript. We gratefully acknowledge these contributors to the study: Cynthia Beaumont, Marie Haverfield, Kristina Kennedy, Rebecca Losh, Camille Mack, Amia Nash, Alexandra Shelley, Emmeline Taylor, and KaSheena Winston; and consultants Barbara McCrady, Gregory Stuart, and Lance Brendan Young.

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